

Ayush Shende

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SUMMARY

Computer Science undergraduate (CGPA 8.95/10) with hands-on experience building and deploying full-stack and mobile applications using Java, Spring Boot, Flutter, and Python. Proficient in REST API design, relational databases, and encryption-backed backend systems. Actively developing skills in data structures, algorithms, and scalable system design; seeking a software engineering internship to contribute to real-world product development.

EDUCATION

Jhulelal Institute of Technology

Bachelor of Technology in Computer Science and Engineering | **CGPA: 8.95/10**

Nagpur, Maharashtra

Aug 2024 – May 2028 (Expected)

EXPERIENCE

JPMorgan Chase (Forge Software Engineering Simulation)

Mar 2026

Software Engineering Virtual Intern

Remote

- Integrated **Apache Kafka** into a Spring Boot microservice to consume and deserialise high-throughput financial transaction messages with a configurable topic and embedded test broker.
- Implemented **transactional data persistence** using Spring Data JPA and H2 SQL — designed entity models, relational schema, and atomic balance updates across User records.
- Exposed a **RESTful account-balance endpoint** via Spring MVC and consumed an external Incentive API (**RestTemplate**) to embed reward logic into the transaction pipeline.
- Verified end-to-end correctness through **Maven unit test suites** following an **Agile** iterative workflow, covering Kafka ingestion, JPA persistence, and third-party API integration.

PROJECTS

UPI Offline Mesh | Java 17, Spring Boot 3, RSA-OAEP, AES-256-GCM, JUnit 5, Docker | [GitHub](#) [Live](#)

2026

- Architected a **mesh-routed deferred payment system** where UPI transactions propagate through offline Bluetooth relay nodes and settle atomically once any node regains connectivity.
- Secured packet transit with **hybrid RSA-OAEP + AES-256-GCM encryption** — relay nodes carry ciphertext they cannot decrypt or modify, enforcing end-to-end confidentiality.
- Achieved **exactly-once settlement** via dual-layer idempotency: SHA-256 fingerprinting with **ConcurrentHashMap.putIfAbsent** plus a DB unique-index guard against concurrent duplicate uploads.
- Authored **3 JUnit 5 unit tests** (round-trip integrity, tamper rejection, concurrent 3-thread duplicate settlement); containerised with **Docker** and deployed live on Render with zero-config startup.

Splitmate — Expense Manager | Flutter, Dart, Firebase Firestore, Firebase Auth | [GitHub](#)

2025

- Developed a **cross-platform mobile application** for shared expense tracking with real-time Firestore sync, group balance calculation, and automated debt-splitting across **multi-member groups**.
- Delivered **monthly analytics**, category-wise spending breakdowns, and on-demand **PDF report generation**; secured with Firebase Authentication and persistent session management.

DPI Engine — Deep Packet Inspection System | C++17, PCAP, TCP/IP, TLS, Multithreading, Winsock2 | [GitHub](#)

2026

- Engineered a **high-performance C++17 network traffic analyser** that parses raw PCAP captures, decodes Ethernet/IPv4/TCP/UDP protocol layers, and extracts **TLS SNI** and **HTTP Host** headers for Layer-7 application identification.
- Implemented **stateful 5-tuple flow tracking** and a configurable rule engine to block connections by application, domain pattern, or IP — forwarding or dropping packets into a filtered output PCAP.
- Architected a **multi-threaded load-balancer + fast-path pipeline** using a thread-safe producer-consumer queue, dispatching packets across parallel fast-path workers for concurrent packet processing.

TECHNICAL SKILLS

Languages: C, C++17, Java, JavaScript, Dart, Python

Frameworks & Libraries: Spring Boot, FastAPI, Flutter, Node.js, Express.js

Databases & Messaging: SQL, Firebase Firestore, SQLite, H2, Apache Kafka, REST APIs

Tools & Practices: Git, Docker, Maven, Agile, Unit Testing

CS Fundamentals: OOP, Data Structures & Algorithms, DBMS, Operating Systems, Computer Networks

Key Concepts: Microservices, Scalable Systems, Hybrid Encryption, Idempotency, Network Protocols (TCP/IP, TLS, HTTP), Multithreading

CERTIFICATIONS & PROFILES

Oracle AI Foundations Associate (2025) | **Google Gen AI Intensive — 5 Day** (2025) | **NPTEL:** Managerial Economics; CSR (2025)

LeetCode: 100+ problems — Arrays, Linked Lists, Binary Search, Sliding Window, Stacks, Dynamic Programming